

## EX8: Spark Tool Installation for the System

### Aim

To install and configure **Apache Spark** on the system and verify its working using PySpark.

### Procedure

#### Step 1: Check Java Installation

Open Command Prompt and type:

```
java -version
```

✓ Ensure Java is installed properly.

#### Step 2: Install Python (for PySpark)

Install: Python 3.10

Check version:

```
python --version
```

#### Step 3: Download Spark

Download: Apache Spark

✓ Choose:

Pre-built for Hadoop

Extract and place in:

C:\spark

#### Step 4: Set Environment Variables

Add System Variable:

SPARK\_HOME = C:\spark

Update PATH

Add the following paths:

C:\spark\bin

Click: java path also

#### Step 5: Start PySpark

Open Command Prompt and type:

```
pyspark
```

✓ If successful, you will see:

SparkSession available as 'spark'

```
>>>
```

#### Step 6: Verify Spark

Run a simple command:

```
sc.parallelize([1,2,3]).collect()
```

#### Step 7: Exit PySpark

```
quit()
```

#### Output

Spark starts successfully

PySpark prompt (>>>) appears

Sample output:

```
[1, 2, 3]
```

#### Result

Thus, **Apache Spark** was successfully installed and verified on the system.

## EX9.1 : RDD Creation and Display Program

### Aim

To create an RDD and display its elements using **Apache Spark**.

### Procedure

Step 1: Open Command Prompt

Step 2: Start PySpark  
pyspark

Step 3: Enter the Program  
data = [10, 20, 30, 40]  
rdd = sc.parallelize(data)  
result = rdd.collect()  
print(result)

Step 4: Execute the Program  
Press **Enter** after each line.

Step 5: Exit PySpark  
quit()

### Output

[10, 20, 30, 40]

### Result

Thus, the RDD was successfully created and displayed using **Apache Spark**.

## EX9.2 : RDD Map Operation in Spark

### Aim

To perform a map operation on an RDD using **Apache Spark**.

### Procedure

**Step 1:** Open Command Prompt

**Step 2:** Start PySpark

```
pyspark
```

**Step 3:** Enter the Program

```
data = [1, 2, 3, 4]
```

```
rdd = sc.parallelize(data)
```

```
result = rdd.map(lambda x: x * 2)
```

```
print(result.collect())
```

**Step 4:** Execute the Program

Press Enter after each line.

**Step 5:** Exit PySpark

```
quit()
```

### Output

```
[2, 4, 6, 8]
```

### Result

Thus, the map operation was successfully performed using **Apache Spark**.

## EX.10: RDD Filter and Word Count Program

### Aim

To perform filter operation and word count using **Apache Spark**.

### Procedure

Step 1: Open Command Prompt

Step 2: Start PySpark

```
pyspark
```

Step 3: **Part A: Filter Operation**

Enter Filter Program

```
data = [10, 15, 20, 25]
rdd = sc.parallelize(data)
result = rdd.filter(lambda x: x > 15)
print(result.collect())
```

Output

```
[20, 25]
```

Step 4: **Part B: Word Count Program**

Enter Word Count Program

```
data = ["hello spark", "hello world"]
rdd = sc.parallelize(data)
words = rdd.flatMap(lambda x: x.split(" "))
count = words.map(lambda x: (x, 1)).reduceByKey(lambda a, b: a + b)
print(count.collect())
```

Output

```
[('hello', 2), ('spark', 1), ('world', 1)]
```

Step 5: Exit PySpark

```
quit()
```

### Result

Thus, filter and word count operations were successfully performed using **Apache Spark**.

